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Inkjet head management apparatus and inkjet printing apparatus

Abstract

An inkjet head management apparatus for managing a humidity of an inkjet head including nozzles for ejecting droplets in an inkjet manner includes a housing having an opening on at least one surface thereof, a closer for opening or closing the opening of the housing, and a dehumidifier disposed in the housing to provide a dehumidifying atmosphere toward the opening when the opening is opened.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION

(1) This application claims the benefit of Korean Patent Application No. 10-2022-0054248, filed on May 2, 2022, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein in its entirety by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to a semiconductor apparatus and, more particularly, to an inkjet head management apparatus and an inkjet printing apparatus.

2. Description of the Related Art

(3) When a printing process is performed on a transparent substrate to manufacture a display device, a printing apparatus including an inkjet head may be used. Color filters, electrodes, etc. of the display may be printed by ejecting droplets through nozzles of the inkjet head. Currently, quantum dots capable of emitting light by using an applied voltage or of emitting light by absorbing ambient light may be printed on a display substrate in the form of ink and used to display colors on the display substrate.

(4) In the process using the inkjet head, the droplets need to be ejected well from a plurality of nozzles included in the inkjet head. When any of the plurality of nozzles does not eject the droplets well, it is regarded as process failure. That is, the nozzle which does not eject the droplets causes the Mura effect and thus is regarded as process failure. As such, when the droplets are not ejected from any nozzle, a maintenance process is performed on the nozzle through high-pressure purging or the like.

(5) To prevent the inkjet nozzles from being clogged, it is considered significant to control the wetness of the inkjet head and the inkjet nozzles. In general, the drier it is, the more likely particles in the ink clog the nozzles. However, the high wetness of the inkjet head and the nozzles is not the only requirement. For example, an inkjet head using ink including a component such as metal or alloy to print electrodes may have a longer lifespan in a dry environment than in a humid environment. Ink including an inorganic component may not realize desired quality because the ink may deteriorate or the inorganic component is oxidized by moisture.

SUMMARY OF THE INVENTION

(6) General inkjet printing apparatuses merely control the humidity of an overall printing process atmosphere and do not specifically control the humidity of an inkjet head part. Particularly, the general inkjet printing apparatuses do not keep only the inkjet head part dry separately from the printing process environment. Furthermore, because the humidity of the inkjet printing process atmosphere may be greatly affected by the season, exposure to an external environment due to accesses of workers to inkjet printing facility, etc., delicate humidity control of the inkjet head part is required. In addition, because special-purpose inkjet heads are high-priced, frequent replacement thereof may increase production costs and cause non-uniform printing quality.

(7) The present invention provides an inkjet head management apparatus and inkjet printing apparatus capable of controlling a dehumidifying or drying atmosphere for an inkjet head.

(8) The present invention also provides an inkjet head management apparatus and inkjet printing apparatus capable of separately maintaining the dehumidifying or drying atmosphere without being directly affected by a printing process atmosphere in an inkjet printing process, and of rapidly providing the dehumidifying atmosphere in a maintenance process of the inkjet head.

(9) The present invention also provides an inkjet head management apparatus and inkjet printing apparatus capable of increasing a lifespan of the inkjet head and of preventing deterioration of ink.

(10) However, the scope of the present invention is not limited thereto.

(11) According to an aspect of the present invention, there is provided an inkjet head management apparatus for managing a humidity of an inkjet head including nozzles for ejecting droplets in an inkjet manner, the inkjet head management apparatus including a housing having an opening on at least one surface thereof, a closer for opening or closing the opening of the housing, and a dehumidifier disposed in the housing to provide a dehumidifying atmosphere toward the opening when the opening is opened.

(12) A cover made of a flame retardant or thermal insulation material may be disposed on at least outer surfaces of the housing.

(13) The opening may be provided on a top surface of the housing, and the inkjet head may approach the top surface of the housing.

(14) The closer may open the opening in a humidity management process of the inkjet head, and close the opening in other processes.

(15) The closer may include a power source mounted in the housing, and a shutter moved from the

opening to a portion other than the opening by an operation of the power source.

(16) The dehumidifier may include a desiccant for providing a dehumidifying atmosphere toward the opening, and a desiccant supporter for supporting the desiccant.

(17) The desiccant supporter may include a desiccant frame, and a mesh connected to the desiccant frame and allowing air to pass therethrough.

(18) The inkjet head management apparatus may further include a heater disposed in the housing to provide heat to the dehumidifier.

(19) The heater may be disposed under the dehumidifier to provide heat to the dehumidifier while the opening is closed.

(20) The inkjet head management apparatus may further include a desiccant weight sensor for measuring a weight of the desiccant in the dehumidifier.

(21) When the weight of the desiccant is greater than a preset value, the heater may provide heat to the desiccant to evaporate moisture of the desiccant and reduce the weight of the desiccant.

(22) The inkjet head management apparatus may further include a flexible contact part mounted around the opening on the housing to absorb impact upon contact with the inkjet head.

(23) The inkjet head management apparatus may further include a humidity sensor disposed in the housing to sense a humidity near the opening.

(24) The inkjet head management apparatus may further include a droplet discharger disposed in the housing to purge and suck droplets from the nozzles.

(25) The inkjet head management apparatus may further include a vent mounted through a side of the housing to maintain an inner space of the housing at atmospheric pressure.

(26) According to another aspect of the present invention, there is provided an inkjet printing apparatus for performing printing on a substrate by ejecting droplets in an inkjet manner, the inkjet printing apparatus including a substrate supporter for disposing the substrate thereon, an inkjet head including nozzles for ejecting droplets onto the substrate, a head mover for moving the inkjet head, and an inkjet head humidity manager for managing a humidity of the inkjet head, wherein the inkjet head humidity manager includes a housing having an opening on at least one surface thereof, a closer for opening or closing the opening of the housing, and a dehumidifier disposed in the housing to provide a dehumidifying atmosphere toward the opening when the opening is opened.

(27) The head mover may move the inkjet head downward on the inkjet head humidity manager to dispose the nozzles under the inkjet head on the opening of the inkjet head humidity manager.

(28) The closer may close the opening when the inkjet head performs printing on the substrate, and open the opening in a humidity management process of the inkjet head.

(29) In the humidity management process of the inkjet head, the dehumidifier may provide a dehumidifying atmosphere to the inkjet head to reduce deterioration or oxidization of droplets caused by moisture in the inkjet head.

(30) According to another aspect of the present invention, there is provided an inkjet head management apparatus for managing a humidity of an inkjet head including nozzles for ejecting droplets in an inkjet manner, the inkjet head management apparatus including a housing having an opening on a top surface thereof to allow the inkjet head to approach, a closer for opening or closing the opening of the housing, a dehumidifier disposed in the housing to provide a dehumidifying atmosphere toward the opening when the opening is opened, and a humidity sensor for sensing a humidity near the opening, wherein the dehumidifier includes a desiccant for providing a dehumidifying atmosphere toward the opening, and a desiccant supporter for supporting the desiccant and including a mesh allowing air to pass therethrough, wherein the closer closes the opening when the inkjet head performs printing on the substrate, and opens the opening in a humidity management process of the inkjet head, and wherein, in the humidity management process of the inkjet head, the dehumidifier provides a dehumidifying atmosphere to the inkjet head to reduce deterioration or oxidization of droplets caused by moisture in the inkjet head.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other features and advantages of the present invention will become more apparent by describing in detail embodiments thereof with reference to the attached drawings in which:
- (2) FIG. 1 is a perspective view of a printing system according to an embodiment of the present invention;
- (3) FIG. 2 is a plan view of a printing system according to an embodiment of the present invention;
- (4) FIG. 3 is a side cross-sectional view of an inkjet head management apparatus according to an embodiment of the present invention;
- (5) FIG. 4 is a side cross-sectional view showing a closed state of an inkjet head management apparatus, according to an embodiment of the present invention;
- (6) FIG. 5 includes plan views showing a procedure of closing an opening of an inkjet head management apparatus, according to an embodiment of the present invention; and
- (7) FIGS. 6 and 7 are side cross-sectional views showing a procedure in which an inkjet head approaches an inkjet head management apparatus and a humidity management process is performed, according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

- (8) Hereinafter, the present invention will be described in detail by explaining embodiments of the invention with reference to the attached drawings.
- (9) The invention may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the concept of the invention to one of ordinary skill in the art. In the drawings, the thicknesses or sizes of layers are exaggerated for clarity and convenience of explanation.
- (10) Embodiments of the invention are described herein with reference to schematic illustrations of idealized embodiments (and intermediate structures) of the invention. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, the embodiments of the invention should not be construed as limited to the particular shapes of regions illustrated herein, but are to include deviations in shapes that result, for example, from manufacturing.
- (11) FIG. 1 is a perspective view of a printing system **100** according to an embodiment of the present invention. FIG. 2 is a plan view of the printing system **100** according to an embodiment of the present invention.
- (12) Referring to FIGS. 1 and 2, the printing system **100** may include a base **110**, a substrate supporter **120**, a gantry **130**, a gantry mover **140**, an inkjet head **150**, a head mover **160**, a droplet ejection amount measurer **170**, a nozzle inspector **180**, and a nozzle manager **190**. In this specification, a first direction **10** may correspond to an x-axis direction, a second direction **20** may correspond to a y-axis direction, and a third direction **30** may correspond to a z-axis direction.
- (13) The base **110** may be provided in a cuboid shape having a certain thickness. The substrate supporter **120** is disposed on a top surface of the base **110**.
- (14) The substrate supporter **120** may have a support plate **121** on which a substrate S is placed. The support plate **121** may be a rectangular plate. A rotary driving member **122** may be connected to a bottom surface of the support plate **121**. The rotary driving member **122** may be a rotary motor. The rotary driving member **122** may rotate the support plate **121** about an axis of rotation perpendicular to the support plate **121**. When the support plate **121** is rotated by the rotary driving member **122**, the substrate S may be rotated by the rotation of the support plate **121**. When a long side direction of cells provided on the substrate S to be coated with droplets faces the second

direction **20**, the rotary driving member **122** may rotate the substrate **S** in such a manner that the long side direction of the cells faces the first direction **10**.

(15) The support plate **121** and the rotary driving member **122** may be linearly moved in the first direction **10** by a linear driving member **123**. The linear driving member **123** may include a slider **124** and a guide member **125**. The rotary driving member **122** may be mounted on a top surface of the slider **124**. The guide member **125** may extend in the first direction **10** in the middle of the top surface of the base **110**. The slider **124** may have an embedded linear motor (not shown), and be linearly moved in the first direction **10** along the guide member **125** by the linear motor (not shown).

(16) The gantry **130** may be provided above a path along which the support plate **121** is moved. The gantry **130** may be spaced apart from the top surface of the base **110** in a vertical direction, and disposed in such a manner that a longitudinal direction thereof is parallel to the second direction **20**.

(17) The gantry mover **140** may linearly move the gantry **130** in the first direction **10**. The gantry mover **140** may include a first moving unit **141** and a second moving unit **142**. The first moving unit **141** may be provided at an end of the gantry **130**, and the second moving unit **142** may be provided at another end of the gantry **130**. The first moving unit **141** may slide along a guide rail **111** provided at a side of the base **110**, and the second moving unit **142** may slide along a guide rail **112** provided at another side of the base **110**, thereby linearly moving the gantry **130** in the first direction **10**.

(18) The inkjet head **150** may be coupled to the gantry **130** by the head mover **160**. The inkjet head **150** may be linearly moved in the longitudinal direction of the gantry **130** (or the second direction **20**) or in the third direction **30** by the head mover **160**. In addition, the inkjet head **150** may rotate about an axis parallel to the first direction **10** with respect to the head mover **160**.

(19) The inkjet head **150** ejects droplets onto the substrate **S**. A plurality of inkjet heads **150** may be provided. Three inkjet heads **150**, e.g., a first head unit **151**, a second head unit **152**, and a third head unit **153**, may be provided but the number of inkjet heads **150** is not limited thereto. The plurality of inkjet heads **150** may be coupled to the gantry **130** in a row in the second direction **20**.

(20) Each inkjet head **150** may include a plurality of nozzles **155** (see FIG. 6) for ejecting droplets, and a nozzle plate **157** on which the plurality of nozzles **155** are provided. The inkjet head **150** may be provided with, for example, 128 or 256 nozzles.

(21) The inkjet head **150** may be provided with piezoelectric elements corresponding to the number of nozzles **155**. Droplet ejection amounts of the plurality of nozzles **155** may be independently adjusted by the control of voltages applied to the piezoelectric elements.

(22) The head mover **160** may be provided for each inkjet head **150**. In the current embodiment, because three inkjet head units **151**, **152**, and **153** are provided, three head movers **160** may be provided to correspond to the number of inkjet heads **150**. Unlike this, a single head mover **160** may be provided and, in this case, the inkjet head units **151**, **152**, and **153** may be moved together rather than individually.

(23) The droplet ejection amount measurer **170** may measure a droplet ejection amount of the inkjet head **150**. The droplet ejection amount measurer **170** may be disposed on the base **110** at a side of the substrate supporter **120**. The droplet ejection amount measurer **170** may measure the amount of droplets ejected from all nozzles **155** (see FIG. 6) of the inkjet head **150**. By measuring the droplet ejection amount of the inkjet head **150**, an error of all nozzles **155** of the inkjet head **150** may be macroscopically checked. That is, when the droplet ejection amount of the inkjet head **150** exceeds a reference value, it may be determined that at least one of the inkjet head **150** has an error.

(24) The inkjet head **150** may be moved in the first and second directions **10** and **20** by the gantry mover **140** and the head mover **160** and positioned above the droplet ejection amount measurer **170**. The head mover **160** may move the inkjet head **150** in the third direction **30** to adjust a vertical distance between the inkjet head **150** and the droplet ejection amount measurer **170**.

(25) The nozzle inspector **180** may check an error of the individual nozzles **155** (see FIG. 6) provided in the inkjet head **150**, through optical inspection. When the measurement result of the droplet ejection amount measurer **170** macroscopically indicates that an unspecified nozzle has an error, the nozzle inspector **180** may inspect all nozzles by checking an error of individual nozzles.

(26) The nozzle inspector **180** may be disposed on the base **110** at a side of the substrate supporter **120**. The inkjet head **150** may be moved in the first and second directions **10** and **20** by the gantry mover **140** and the head mover **160** and positioned above the nozzle inspector **180**. The head mover **160** may move the inkjet head **150** in the third direction **30** to adjust a vertical distance between the inkjet head **150** and the nozzle inspector **180**.

(27) The nozzle manager **190** may perform a management process to allow droplets to be readily ejected from the nozzles **155** of the inkjet head **150**. The management process includes a process of adjusting a humidity of the inkjet head **150** or providing a drying/dehumidifying atmosphere. The nozzle manager **190** may be disposed on the base **110** at a side of the substrate supporter **120**. The inkjet head **150** may be moved in the first and second directions **10** and **20** by the gantry mover **140** and the head mover **160** and positioned above the nozzle manager **190**. The head mover **160** may move the inkjet head **150** in the third direction **30** to adjust a vertical distance between the inkjet head **150** and the nozzle manager **190**.

(28) The inkjet heads **150** including the inkjet head units **151**, **152**, and **153** may be moved individually and connected to different head movers **160** or the gantry mover **140**. In this case, while a specific one of the inkjet head units **151**, **152**, and **153** is positioned above the nozzle manager **190** and a humidity management process is being performed, a printing process may be performed on the substrate **S** by the others of the inkjet head units **151**, **152**, and **153**. As such, the printing process and the management process of the inkjet heads **150** may be performed simultaneously and thus a process time may be reduced.

(29) Meanwhile, the printing system **100** may further include a droplet supply device **210**. The droplet supply device **210** may be mounted on and at a side of the gantry **130**. The droplet supply device **210** includes a droplet supply module **211** and a pressure adjustment module **212**. The droplet supply module **211** and the pressure adjustment module **212** may be coupled to the gantry **130**. The droplet supply module **211** receives droplets from the droplet supply device **210**, and supplies the droplets to the inkjet head **150**. The pressure adjustment module **212** adjusts a pressure of the droplet supply module **211** by providing a positive or negative pressure to the droplet supply module **211**.

(30) FIG. 3 is a side cross-sectional view of an inkjet head management apparatus **300** according to an embodiment of the present invention. The inkjet head management apparatus **300** corresponds to the nozzle manager **190** of FIGS. 1 and 2. In relation to FIG. 3 and the subsequent drawings, the nozzle manager **190** will be referred to as or used interchangeably with the inkjet head management apparatus **300**.

(31) Referring to FIG. 3, the inkjet head management apparatus **300** may be provided to manage a humidity of the inkjet head **150**. The inkjet head management apparatus **300** may be provided to manage the humidity only for the inkjet head **150** among various elements of the inkjet printing system **100**. Particularly, the inkjet head management apparatus **300** may be provided to control a dehumidifying or drying atmosphere for the inkjet head **150**.

(32) The inkjet head management apparatus **300** may include a housing **310**, a closer **320**, and a dehumidifier **330**. According to an embodiment, the inkjet head management apparatus **300** may further include a heater **340**, a desiccant weight sensor **350**, a flexible contact part **360**, a humidity sensor **370**, a droplet discharger **380**, a vent **390**, and a controller **400**.

(33) The housing **310** may be provided as a body of the inkjet head management apparatus **300** to dispose various elements of the inkjet head management apparatus **300** in its inner space **311**. The housing **310** may have a substantially hexahedral shape but is not limited thereto.

(34) The housing **310** may have an opening **315** on at least one surface thereof. The opening **315** is

open to allow the inner space **311** of the housing **310** to interact with an atmosphere outside the housing **310**. The opening **315** is provided to allow the inkjet head **150** to approach. When the inkjet head **150** approaches, it means that the inkjet head **150** or the nozzles **155** are brought into close contact with, inserted into, or disposed close to the opening **315**.

(35) The opening **315** may be provided on a top surface of the housing **310** such that the nozzles **155** (see FIG. 6) of the inkjet head **150** may move downward to approach. However, the opening **315** may be provided on another surface of the housing **310** as long as the opening **315** allows the inkjet head **150** to approach. The opening **315** may have a size corresponding to the size of one or more head units **151**, **152**, and **153**.

(36) According to an embodiment, a cover **317** may be disposed on at least outer surfaces of the housing **310**. The cover **317** may be provided as a separate plate connected or attached to the outer surfaces of the housing **310**, or integrated with the outer surfaces of the housing **310**. The cover **317** may be mounted to prevent an atmosphere of the inner space **311** of the housing **310** from affecting elements of the printing system **100** outside the housing **310**. For example, the cover **317** may prevent heat energy of the inner space **311** of the housing **310** from being transferred to the other elements of the printing system **100**. The cover **317** may be made of a flame retardant or thermal insulation material such as expanded polystyrene or glass fiber.

(37) The closer **320** is provided to open or close the opening **315** of the housing **310**. According to an embodiment, the closer **320** may include a power source **321** and a shutter **325**.

(38) The power source **321** may be mounted in the housing **310** to provide power for moving the shutter **325**. The shutter **325** may directly open or close the opening **315**. The shutter **325** may be moved from the opening **315** to a portion other than the opening **315** or vice versa by the operation of the power source **321**. A power transmitter **323** for transmitting power may be further interposed between the power source **321** and the shutter **325**. For example, the power source **321** may include a motor and the shutter **325** may include a plurality of unit shutters, and the shutter **325** may be pushed or pulled by the operation of the motor to open or close the opening **315**. Alternatively, the shutter **325** may be wound or unwound by the operation of the power source **321** to open or close the opening **315**.

(39) The dehumidifier **330** may be disposed in the housing **310** to provide a dehumidifying or drying atmosphere toward the opening **315**. The dehumidifier **330** may provide a dehumidifying or drying atmosphere toward the opening **315** when the opening **315** is opened. Because the inkjet head **150** is close to the opening **315** when the opening **315** is opened, the dehumidifier **330** may provide a dehumidifying or drying atmosphere to the inkjet head **150**.

(40) The dehumidifier **330** may include a desiccant **331** and a desiccant supporter **333** and **335**.

(41) Any material having a dehumidification function may be used as the desiccant **331** without limitation. For example, silica gel may be used as the desiccant **331**. Silica gel is a porous granular material containing silicon dioxide, i.e., silica, as a main component, and has a large specific surface area due to its porous structure and may absorb moisture well. In addition, silica gel is easily reusable because the absorbed moisture is released when heat is applied.

(42) The desiccant supporter **333** and **335** may include a desiccant frame **333** and a mesh **335**. The desiccant frame **333** may be connected to the inside of the housing **310** or the droplet discharger **380**. The desiccant frame **333** may be provided detachably. The desiccant frame **333** may have a hollow hole to which the mesh **335** is connected. The desiccant frame **333** may be made of a rigid material such as metal to support the weights of the mesh **335** and the desiccant **331** contained in the mesh **335**.

(43) The mesh **335** is provided to contain the desiccant **331**. The mesh **335** has a mesh structure through which air passes well, to allow the supported desiccant **331** to easily absorb or release moisture. The mesh **335** may have an opening size less than a diameter of the desiccant **331** to prevent the desiccant **331** from leaking downward.

(44) Meanwhile, the dehumidifier **330** may not consist of the desiccant **331** and the desiccant

supporter **333** and **335**, but be provided as an electric dehumidifier for performing a dehumidification function.

(45) The heater **340** may be disposed in the housing **310** to provide heat to the inner space **311** of the housing **310**. Particularly, the heat of the heater **340** may be provided to the dehumidifier **330**. The heater **340** may be provided as a hot-wire coil heater, or any means capable of generating heat, e.g., a plate-type heater, may also be used without limitation. The moisture absorbed by the desiccant **331** of the dehumidifier **330** may be released by the heat applied by the heater **340**. The desiccant **331** dried by the heat provided from the heater **340** may provide a dehumidifying or drying atmosphere toward the opening **315** again.

(46) The heater **340** may be disposed under the dehumidifier **330**. When the heater **340** is operated, the inner space **311** of the housing **310**, and more particularly, the desiccant **331** may be in a high-humidity status. The heat generated by the heater **340** is transferred upward through convection, radiation, or the like, and the heat may be efficiently transferred to the dehumidifier **330** disposed on the heater **340**. The heater **340** may provide heat to the dehumidifier **330** after a humidity management process for providing a dehumidifying or drying atmosphere to the inkjet head **150** is finished. That is, the heater **340** may provide heat while the opening **315** is closed by the closer **320**.

(47) The dehumidifier **330** may further include the desiccant weight sensor **350**. The desiccant weight sensor **350** may be mounted on the desiccant frame **333** or the mesh **335**. The desiccant weight sensor **350** may measure a weight of the desiccant **331** in real time. According to an embodiment, when it is determined that the weight of the desiccant **331** is increased by a specific rate or more compared to an initial weight of the desiccant **331**, the desiccant **331** may be replaced.

(48) According to another embodiment, when it is determined that the weight of the desiccant **331** is increased by a specific rate or more compared to an initial weight of the desiccant **331**, the weight of the desiccant **331** may be reduced by operating the heater **340** to evaporate moisture of the desiccant **331**. When the desiccant **331** is dried by evaporating moisture, a period of use may be extended. As such, the desiccant **331** may be reused.

(49) The flexible contact part **360** may be further provided on the housing **310**. The flexible contact part **360** may be mounted around the opening **315**. The flexible contact part **360** may use an O-ring, or any flexible means such as a rubber or elastic material may also be used without limitation. The flexible contact part **360** may be mounted around the opening **315** to minimize contact impact with the housing **310** when the inkjet head **150** approaches. Because the inkjet head **150** and the housing **310** are made of hard materials and thus may be damaged when brought into contact with each other, the flexible contact part **360** may absorb contact impact between the inkjet head **150** and the housing **310**. In addition, the flexible contact part **360** may seal a gap to achieve close contact between the inkjet head **150** and the housing **310**.

(50) The humidity sensor **370** may be disposed in the housing **310** to sense a humidity of the inner space **311**. Particularly, the humidity sensor **370** may be provided to sense a humidity near the opening **315**. Based on the humidity information sensed by the humidity sensor **370**, it may be determined whether the inkjet head management apparatus **300** is required to perform a humidity management process on the inkjet head **150**. When the humidity is high, the heater **340** may be operated to reduce the humidity of the dehumidifier **330** or the inner space **311**.

(51) The droplet discharger **380** may be disposed in the housing **310** to purge and suck droplets from the nozzles **155** of the inkjet head **150**. Droplets may remain at ends of the nozzles **155** after a printing process on the substrate **S** is finished. These droplets may deteriorate or be contaminated while the inkjet head **150** is moving in the printing system **100**, and process failure may be caused when a subsequent printing process is performed while the droplets remain. Alternatively, when some nozzles are clogged with the droplets, the droplets may not be ejected well and thus process failure may be caused in a printing process. Therefore, a purger **381** may be provided to collect the droplets in the procedure of discharging the droplets remaining at the ends of the nozzles **155**. The

droplets falling into the purger **381** may be discharged to an external droplet processing means (not shown). A sucker **385** may suck the droplets remaining in the nozzles **155**, by approaching the nozzles **155** and providing a suction pressure. When the nozzles **155** are clogged, the sucker **385** may unclog the nozzles **155** by using the suction pressure. A level of the suction pressure of the sucker **385** may be controlled by opening or closing a suction valve **387**. The droplets sucked into the sucker **385** may be discharged to an external droplet processing means (not shown).

(52) The vent **390** may be mounted through a side of the housing **310**. The vent **390** may be provided to match a pressure of the inner space **311** of the housing **310** with an external pressure. An amount of external air introduced into the housing **310** may be controlled by opening or closing a vent valve **395** of the vent **390**. A filter (not shown) for filtering out fine particles when external air is introduced may be further mounted on the vent **390**.

(53) For example, a negative pressure may be formed in the inner space **311** in the procedure of purging and sucking a waste liquid of the inkjet head **150**. Because the droplets in the nozzles **155** may flow due to the negative pressure, the vent **390** may introduce the external air to maintain the inner space **311** of the housing **310** at atmospheric pressure. As another example, moisture evaporated by heating the dehumidifier **330** by the heater **340** may escape to the outside through the vent **390** not to remain in the inner space **311** of the housing **310**.

(54) The controller **400** may control a series of operations of the inkjet head management apparatus **300**. For example, the controller **400** may control operation of the closer **320** in consideration of positions of the inkjet head **150** and the opening **315**. As another example, the controller **400** may transmit a notification for replacing the desiccant **331**, to a user based on the weight measurement data of the desiccant **331** sensed by the desiccant weight sensor **350**, or control the heater **340** to operate to evaporate the moisture of the desiccant **331**. As another example, the controller **400** may determine whether the inkjet head management apparatus **300** is appropriate to perform a humidity management process on the inkjet head **150**, based on the humidity information sensed by the humidity sensor **370**. As another example, the controller **400** may control opening or closing of the suction valve **387** or the vent valve **395**.

(55) FIG. **4** is a side cross-sectional view showing a closed state of the inkjet head management apparatus **300**, according to an embodiment of the present invention. FIG. **5** includes plan views showing a procedure of closing the opening **315** of the inkjet head management apparatus **300**, according to an embodiment of the present invention.

(56) Referring to FIGS. **4** and **5**, the opening **315** of the housing **310** may be closed by the closer **320**. The power source **321** may provide power to move the shutter **325** to the opening **315** from a position other than the opening **315**.

(57) Referring to the first view of FIG. **5**, the opening **315** is open. The power source **321** may provide power to position the shutter **325** at a position other than the opening **315**. This state may be a state immediately before the inkjet head **150** approaches the opening **315**. Alternatively, this state may be a state in which the inkjet head **150** leaves the opening **315** immediately after a humidity management process of the inkjet head **150** is finished.

(58) Referring to the second view of FIG. **5**, the opening **315** may be partially closed. The power source **321** may provide power to position the shutter **325** at the opening **315**, and the shutter **325** may move to the opening **315** to gradually close the opening **315**.

(59) Referring to the third view of FIG. **5**, the opening **315** is completely closed. The power source **321** may provide power to position the shutter **325** at a portion corresponding to the opening **315**. This state may be a state in which a dehumidifying or drying atmosphere is prepared by controlling a humidity of the inner space **311** of the housing **310** after a humidity management process of the inkjet head **150** is finished. Alternatively, this state may be a state in which the desiccant **331** is replaced or in which moisture contained in the desiccant **331** is evaporated by operating the heater **340**.

(60) In other words, the closer **320** may open the opening **315** only in a humidity management

process of the inkjet head **150**, and close the opening **315** in other processes. The procedure of FIG. **5** may also be performed in reverse order.

(61) Referring back to FIG. **4**, the housing **310** may provide the inner space **311** and the opening **315** may be closed by the closer **320**. The inner space **311** of the housing **310** may be closed separately from an external environment.

(62) Therefore, the inkjet head management apparatus **300** of the present invention may maintain a dehumidifying or drying atmosphere separately from the other elements of the printing system **100** without being directly affected by a printing process atmosphere in an inkjet printing process or by an external process atmosphere. In addition, because a dehumidifying atmosphere may be previously maintained in the inner space **311** of the housing **310**, the dehumidifying atmosphere may be rapidly provided in a maintenance process of the inkjet head **150**. That is, while previously maintaining the dehumidifying atmosphere in the inner space **311** of the housing **310** by closing the opening **315**, when the inkjet head **150** approaches the opening **315**, the opening **315** may be opened and the inkjet head **150** may be dehumidified or dried immediately.

(63) FIGS. **6** and **7** are side cross-sectional views showing a procedure in which the inkjet head **150** approaches the inkjet head management apparatus **300** and a humidity management process is performed, according to an embodiment of the present invention.

(64) Initially, referring to FIG. **6**, the inkjet head **150** may approach the opening **315**. The opening **315** is open because the shutter **325** is moved into the housing **310**. The inkjet head **150** may be moved in the first and second directions **10** and **20** by the head mover **160** (and the gantry mover **140**) and positioned above the inkjet head management apparatus **300** (or the nozzle manager **190**). Then, the head mover **160** may move the inkjet head **150** downward in the third direction **30** to approach the opening **315**. The inkjet head **150** may be brought into close contact with the opening **315** and, in this case, the flexible contact part **360** may be interposed between the inkjet head **150** and the opening **315** to absorb contact impact.

(65) While the inkjet head **150** is close to the opening **315**, the dehumidifier **330** may provide a dehumidifying or drying atmosphere DH to the inkjet head **150**. Because a space between the dehumidifier **330** and the opening **315** is already maintained in an atmosphere with a lower humidity than the outside of the inkjet head management apparatus **300**, the nozzles **155** of the inkjet head **150** having approached the opening **315** may be reduced in humidity. That is, ends of the nozzles **155** may be dehumidified or dried and thus droplets included in the nozzles **155** may be prevented from deteriorating or being oxidized by moisture. Particularly, inorganic components of the droplets may be prevented from deteriorating or being oxidized by moisture.

(66) A process of purging and sucking droplets from the nozzles **155** by operating the droplet discharger **380** may also be performed while the inkjet head **150** is close to the opening **315**.

(67) Then, referring to FIG. **7**, after the humidity management process of the inkjet head **150** is finished, the head mover **160** may move the inkjet head **150** upward in the third direction **30** away from the opening **315**. At the same time, the shutter **325** may move to the opening **315** to close the opening **315**.

(68) The nozzles **155** of the inkjet head **150** have gone through the management process including dehumidification, purging, and suction. The inkjet head **150** may be moved in the first and second directions **10** and **20** by the head mover **160** (and the gantry mover **140**) and used for a printing process on the substrate S.

(69) In addition, the inkjet head management apparatus **300** may operate to reduce the humidity of the inner space **311** again and maintain a dehumidifying or drying atmosphere. For example, the desiccant weight sensor **350** may measure a weight of the desiccant **331** and, when the weight of the desiccant **331** is greater than a preset value, the heater **340** may operate and provide heat H to evaporate moisture of the desiccant **331**. The vent **390** may open the vent valve **395** to release the moisture evaporated from the desiccant **331**, to the outside and, at the same time, maintain the inner space **311** at atmospheric pressure.

(70) As described above, according to the inkjet head management apparatus 300 of the present invention, a lifespan of an inkjet head may be increased and deterioration of ink may be prevented by controlling a dehumidifying or drying atmosphere for the inkjet head. Furthermore, the dehumidifying or drying atmosphere may be separately maintained without being directly affected by a printing process atmosphere in an inkjet printing process, and the dehumidifying atmosphere may be rapidly provided in a maintenance process of the inkjet head.

(71) As described above, according to an embodiment of the present invention, a dehumidifying or drying atmosphere may be controlled for an inkjet head.

(72) In addition, according to an embodiment of the present invention, the dehumidifying or drying atmosphere may be separately maintained without being directly affected by a printing process atmosphere in an inkjet printing process, and the dehumidifying atmosphere may be rapidly provided in a maintenance process of the inkjet head.

(73) Furthermore, according to an embodiment of the present invention, a lifespan of the inkjet head may be increased and deterioration of ink may be prevented.

(74) However, the scope of the present invention is not limited to the above effects.

(75) While the present invention has been particularly shown and described with reference to embodiments thereof, it will be understood by one of ordinary skill in the art that various changes in form and details may be made therein without departing from the scope of the present invention as defined by the following claims.

Claims

1. An inkjet head management apparatus for managing a humidity of an inkjet head comprising nozzles for ejecting droplets in an inkjet manner, the inkjet head management apparatus comprising: a housing having an opening on at least one surface thereof; a closer for opening or closing the opening of the housing; and a dehumidifier disposed in the housing to provide a dehumidifying atmosphere toward the opening when the opening is opened, wherein the dehumidifier comprises: a desiccant for providing a dehumidifying atmosphere toward the opening, and a desiccant supporter for supporting the desiccant, and wherein the desiccant supporter comprises a desiccant frame, and a mesh connected to the desiccant frame and allowing air to pass therethrough.
2. The inkjet head management apparatus of claim 1, wherein a cover made of a flame retardant or thermal insulation material is disposed on at least outer surfaces of the housing.
3. The inkjet head management apparatus of claim 1, wherein the opening is provided on a top surface of the housing, and the inkjet head approaches the top surface of the housing.
4. The inkjet head management apparatus of claim 3, further comprising a flexible contact part mounted around the opening on the housing to absorb impact upon contact with the inkjet head.
5. The inkjet head management apparatus of claim 1, wherein the closer opens the opening in a humidity management process of the inkjet head, and closes the opening in other processes.
6. The inkjet head management apparatus of claim 1, wherein the closer comprises: a power source mounted in the housing; and a shutter moved from the opening to a portion other than the opening by an operation of the power source.
7. The inkjet head management apparatus of claim 1, further comprising a heater disposed in the housing to provide heat to the dehumidifier.
8. The inkjet head management apparatus of claim 7, wherein the heater is disposed under the dehumidifier to provide heat to the dehumidifier while the opening is closed.
9. The inkjet head management apparatus of claim 7, further comprising a desiccant weight sensor for measuring a weight of the desiccant in the dehumidifier.
10. The inkjet head management apparatus of claim 9, wherein, when the weight of the desiccant is greater than a preset value, the heater provides heat to the desiccant to evaporate moisture of the

desiccant and reduce the weight of the desiccant.

11. The inkjet head management apparatus of claim 1, further comprising a humidity sensor disposed in the housing to sense a humidity near the opening.

12. The inkjet head management apparatus of claim 1, further comprising a droplet discharger disposed in the housing to purge and suck droplets from the nozzles.

13. The inkjet head management apparatus of claim 12, further comprising a vent mounted through a side of the housing to maintain an inner space of the housing at atmospheric pressure.

14. An inkjet printing apparatus for performing printing on a substrate by ejecting droplets in an inkjet manner, the inkjet printing apparatus comprising: a substrate supporter for disposing the substrate thereon; an inkjet head comprising nozzles for ejecting droplets onto the substrate; a head mover for moving the inkjet head; and an inkjet head humidity manager for managing a humidity of the inkjet head, wherein the inkjet head humidity manager comprises: a housing having an opening on at least one surface thereof; a closer for opening or closing the opening of the housing; and a dehumidifier disposed in the housing to provide a dehumidifying atmosphere toward the opening when the opening is opened, wherein the dehumidifier comprises: a desiccant for providing a dehumidifying atmosphere toward the opening, and a desiccant supporter for supporting the desiccant, and wherein the desiccant supporter comprises a desiccant frame, and a mesh connected to the desiccant frame and allowing air to pass therethrough.

15. The inkjet printing apparatus of claim 14, wherein the head mover moves the inkjet head downward on the inkjet head humidity manager to dispose the nozzles under the inkjet head on the opening of the inkjet head humidity manager.

16. The inkjet printing apparatus of claim 14, wherein the closer closes the opening when the inkjet head performs printing on the substrate, and opens the opening in a humidity management process of the inkjet head.

17. The inkjet printing apparatus of claim 16, wherein, in the humidity management process of the inkjet head, the dehumidifier provides a dehumidifying atmosphere to the inkjet head to reduce deterioration or oxidization of droplets caused by moisture in the inkjet head.

18. An inkjet head management apparatus for managing a humidity of an inkjet head comprising nozzles for ejecting droplets in an inkjet manner, the inkjet head management apparatus comprising: a housing having an opening on a top surface thereof to allow the inkjet head to approach; a closer for opening or closing the opening of the housing; a dehumidifier disposed in the housing to provide a dehumidifying atmosphere toward the opening when the opening is opened; and a humidity sensor for sensing a humidity near the opening, wherein the dehumidifier comprises a desiccant for providing a dehumidifying atmosphere toward the opening; and a desiccant supporter for supporting the desiccant and comprising a mesh allowing air to pass therethrough, wherein the closer closes the opening when the inkjet head performs printing on the substrate, and opens the opening in a humidity management process of the inkjet head, and wherein, in the humidity management process of the inkjet head, the dehumidifier provides a dehumidifying atmosphere to the inkjet head to reduce deterioration or oxidization of droplets caused by moisture in the inkjet head.
